

12.679

EE25BTECH11042 - Nipun Dasari

October 3, 2025

Question

If the area of a triangle with the vertices $\vec{A}(k, 0)$, $\vec{B}(2, 0)$ and $\vec{C}(0, -2)$ is 2 square units, find the value of k .

Theoretical Solution

Given the vertices:

$$\vec{A} = \begin{pmatrix} k \\ 0 \end{pmatrix}, \vec{B} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \vec{C} = \begin{pmatrix} 0 \\ -2 \end{pmatrix} \quad (0.1)$$

We define two side vectors originating from vertex $\vec{C}$.

Vector $\vec{a}$ (from $\vec{C}$ to $\vec{B}$):

$$\vec{a} = \vec{B} - \vec{C} = \begin{pmatrix} 2 - 0 \\ 0 - (-2) \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (0.2)$$

Vector $\vec{b}$ (from $\vec{C}$ to $\vec{A}$):

$$\vec{b} = \vec{A} - \vec{C} = \begin{pmatrix} k - 0 \\ 0 - (-2) \end{pmatrix} = \begin{pmatrix} k \\ 2 \end{pmatrix} \quad (0.3)$$

Theoretical Solution

The area is given by $\Delta = \frac{1}{2} \|\vec{b} \times \vec{a}\|$.

Since the area is 2, we have:

$$\frac{1}{2} \|\vec{b} \times \vec{a}\| = 2 \implies \|\vec{b} \times \vec{a}\| = 4 \quad (0.4)$$

To compute the cross product, we represent the vectors in 3D space:

$$\vec{b} = \begin{pmatrix} k \\ 2 \\ 0 \end{pmatrix} \quad \vec{a} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} \quad (0.5)$$

Theoretical Solution

Using the formula $\vec{A} \times \vec{B} = \begin{pmatrix} |\vec{A}_{23} & \vec{B}_{23}| \\ |\vec{A}_{31} & \vec{B}_{31}| \\ |\vec{A}_{12} & \vec{B}_{12}| \end{pmatrix}$:

$$|\vec{b}_{23} \ \vec{a}_{23}| = \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} = 0 \quad (0.6)$$

$$|\vec{b}_{31} \ \vec{a}_{31}| = \begin{vmatrix} 0 & 0 \\ k & 2 \end{vmatrix} = 0 \quad (0.7)$$

$$|\vec{b}_{12} \ \vec{a}_{12}| = \begin{vmatrix} k & 2 \\ 2 & 2 \end{vmatrix} = 2k - 4 \quad (0.8)$$

The cross product vector is:

$$\vec{b} \times \vec{a} = \begin{pmatrix} 0 \\ 0 \\ 2k - 4 \end{pmatrix} \quad (0.9)$$

The norm is $\|\vec{b} \times \vec{a}\| = \sqrt{0^2 + 0^2 + (2k - 4)^2} = |2k - 4|$.

Theoretical Solution

We now solve for k using the area condition:

$$|2k - 4| = 4 \quad (0.10)$$

This absolute value equation yields two possibilities:

Case 1:

$$2k - 4 = 4 \implies 2k = 8 \implies k = 4 \quad (0.11)$$

Case 2:

$$2k - 4 = -4 \implies 2k = 0 \implies k = 0 \quad (0.12)$$

The possible values for k are 0 or 4.

Plot of Both Solutions

